Beyond planted strings: theory for the VOUCH extension

Working note, not for circulation

Quang-Vinh Dang

This note collects the new results the extension rests on. Notation follows the VOUCH paper: a committed cohort of m pairs, pair contents $C_i = (c_i^0, c_i^1)$, inclusion coins b_i, all remaining randomness of training and unlearning ξ, a score s in the declared class $\mathcal{F}$, signs $z_i = \mathbf{1}[s(M, c_i^{b_i}) > s(M, c_i^{1-b_i})]$ (ties split by a committed coin), and the realised cohort advantage $\Delta = \frac{2}{m}\sum_i z_i - 1$. VOUCH certifies Δ exactly (its Theorem 2) and reaches the coin-averaged advantage through its Proposition 3. The four results below extend that ladder in three directions.

1 Setting and the canary generator

Assumption 1 (Generator)**.** The pair contents $C_1, \ldots, C_m$ are i.i.d. draws from a generator distribution P on ordered pairs that is *label-symmetric*: $(c^0, c^1) \sim P$ implies $(c^1, c^0) \sim P$. The coins b_i are i.i.d. Bern($\frac{1}{2}$), independent of C and of ξ; ξ is independent of C.

Both properties are under the auditor's control: they describe the auditor's own sampler, not the data or the model.

2 Semantic twins keep the exact null

VOUCH's secrets are random strings, which have no paraphrase. A semantic twin pair instead states two incompatible facts about one fictitious entity: c^0 says the entity has attribute value v, c^1 says it has value v', with $\{v, v'\}$ drawn from a value set. The concern is that semantics might break the symmetry that makes the VOUCH null exact. It does not, provided the generator is label-symmetric.

Proposition 2 (Exact null for semantic twins)**.** *Under Assumption 1, if the deployed model is conditionally independent of the coins given the contents (exact unlearning), then for every score s and every step t of a reveal order chosen independently of the signs, $Z_{\pi(t)} \mid \mathcal{F}_{t-1} \sim \mathrm{Bern}(\frac{1}{2})$ exactly. This holds in particular for* belief scores *that compare the two values under the same question, $s(M, c^k) = \log P_M(v_k \mid q)$ averaged over a committed set of paraphrased questions q.*

Proof. The proof of VOUCH Theorem 1 uses only that, given the unordered pair opened at step t, the two scores are fixed numbers and the coin deciding which is the in-twin's is a fair coin independent of everything revealed earlier and of the model. Neither the semantics of c^0, c^1 nor the form of s enters, as long as $s(M, \cdot)$ is a function of the model and the record alone. A belief score of the stated form is such a function: both twins share the question set and differ only in the value evaluated. □

Remark 3 (Power, not validity, is what semantics costs). A base model may prefer one value a priori ("born in Paris" over "born in Porto"). That preference enters both arms symmetrically

and leaves the null exact, but it adds variance to D_i and costs power. The base-calibrated score $s(M, c) - s(M_0, c)$, with M_0 the pre-fine-tuning model, removes it, and remains in $\mathcal{F}$ because M_0 is fixed before the coins are drawn.

3 From the cohort to the generator population

VOUCH Proposition 3 averages over the coins with the contents held fixed. The auditor also controls the contents, through Assumption 1, so the same argument extends to them.

Assumption 4 (Bounded influence)**.** For fixed ξ, changing a single coin b_i changes at most κ of the m signs, and replacing the contents C_i of a single pair changes at most ℓ of them.

Under exact unlearning $\kappa = \ell = 1$: the model carries no trace of the forget set, so a pair's coin or contents can only affect that pair's own sign.

Theorem 5 (Transfer to the generator population)**.** *Let $\Delta_\xi = \mathbb{E}_{b,C}[\Delta \mid \xi]$ be the expected advantage of this pipeline run over fresh cohorts from P. Under Assumptions 1 and 4, for any $\alpha' \in (0, 1)$, with probability at least $1 - \alpha'$ over (b, C),*

$$|\Delta - \Delta_\xi| \;\le\; u_m(\alpha') \;:=\; \sqrt{\frac{2(\kappa^2 + \ell^2)\log(2/\alpha')}{m}}.$$

Consequently a cohort certificate at tolerance ε and level α implies $|\Delta_\xi| < \varepsilon + u_m(\alpha')$ at level $\alpha + \alpha'$.

Proof. Given ξ, Δ is a function of the $2m$ independent coordinates $(b_1, \ldots, b_m, C_1, \ldots, C_m)$. By Assumption 4, changing one coin changes Δ by at most $2\kappa/m$ and changing one pair's contents by at most $2\ell/m$. McDiarmid's inequality gives $\Pr[|\Delta - \Delta_\xi| \ge u] \le 2\exp\bigl(-2u^2/\sum_k c_k^2\bigr)$ with $\sum_k c_k^2 = 4(\kappa^2+\ell^2)/m$, that is $2\exp\bigl(-u^2 m/(2(\kappa^2+\ell^2))\bigr)$; setting it to α' gives u_m. The consequence follows from a union bound with the certificate's own error. □

Remark 6 (What is still not reached)*.* Δ_ξ averages over cohorts but conditions on the training randomness ξ: it is the advantage of *this* deployed pipeline run on the generator population. Averaging over ξ would need several independent training runs and is not attempted. For an auditor of a deployed model, Δ_ξ is the relevant target. At $m = 512$, $\alpha' = 0.05$ and $\kappa = \ell = 1$ the inflation is 0.17, against 0.12 for the coin-only transfer of VOUCH Proposition 3.

4 From canaries to organic records, and the cost of special-casing

Let P_o be a distribution of *organic* record pairs: two records of the kind a user actually deletes, with a hypothetical fair coin choosing which one was trained on. For a pair content c, write

$$h_\xi(c) \;=\; \mathbb{E}\bigl[\,2Z_i - 1 \;\big|\; C_i = c,\ \xi\,\bigr],$$

the expected sign contribution of a pair with contents c placed in the cohort, the expectation being over the coins and over the other pairs' contents. Then $\Delta_\xi = \mathbb{E}_{c\sim P}[h_\xi(c)]$, and we define the *organic advantage* $\Delta_\xi^o = \mathbb{E}_{c\sim P_o}[h_\xi(c)]$: the advantage the same pipeline run would show on an organic pair planted in the same way.

Definition 7 (Distinguishing advantage)**.** For a class $\mathcal{H}$ of functions from pair contents to $[0, 1]$, $\gamma_{\mathcal{H}}(P, P_o) = \sup_{\phi\in\mathcal{H}}\bigl|\mathbb{E}_P\phi - \mathbb{E}_{P_o}\phi\bigr|$. With $\mathcal{H}$ all measurable functions this is the total variation distance.

Theorem 8 (Special-casing requires detection)**.** *Let $\phi_\xi = (1 + h_\xi)/2$, which takes values in $[0, 1]$. Then*

$$|\Delta^o_\xi - \Delta_\xi| \;=\; 2\left|\mathbb{E}_P\phi_\xi - \mathbb{E}_{P_o}\phi_\xi\right| \;\le\; 2\,\gamma_{\mathcal{H}}(P, P_o)$$

for any class $\mathcal{H}$ containing ϕ_ξ. Equivalently: if the pipeline treats canaries and organic records differently enough to open a gap g between their advantages, then ϕ_ξ is a test that tells canary pairs from organic pairs with advantage at least $g/2$.

Proof. $\Delta_\xi - \Delta^o_\xi = \mathbb{E}_P h_\xi - \mathbb{E}_{P_o} h_\xi = 2(\mathbb{E}_P\phi_\xi - \mathbb{E}_{P_o}\phi_\xi)$ by linearity, and the right-hand side is at most $2\gamma_{\mathcal{H}}$ whenever $\phi_\xi \in \mathcal{H}$. □

Remark 9 (How to read Theorem 8). The theorem is a reduction, not a free guarantee. ϕ_ξ is computable by anyone who can run the provider's pipeline on a pair and score both twins, so the relevant class $\mathcal{H}$ is "tests the provider can implement". It turns the honest-but-verifiable assumption of VOUCH into a quantitative one: a provider who special-cases the cohort must, in effect, operate a canary detector, and the gap it can open is at most twice that detector's advantage. Three consequences follow. (i) If the canary list is committed but hidden until after deployment, the provider can only detect canaries from their contents, which is exactly what γ measures. (ii) An auditor cannot certify γ from data: any detector we train gives a *lower* bound on γ, never an upper bound. We therefore report the strongest detectors we can build as evidence, and state the organic claim conditionally on γ. (iii) Making P close to P_o (natural-language canaries in the user's own format) shrinks γ for every detector at once.

Corollary 10 (The full ladder)**.** *Under the assumptions of VOUCH Theorem 2, Theorem 5 and Theorem 8, if a two-sided (ε, α) cohort certificate is issued, then with probability at least $1 - \alpha - \alpha'$,*

$$|\Delta^o_\xi| \;<\; \varepsilon \;+\; u_m(\alpha') \;+\; 2\gamma_{\mathcal{H}}(P, P_o).$$

Each term has a separate cause: the tolerance the auditor chose, the size of the cohort, and how far the canaries are from the data they stand in for.

5 How fast the cohort certificate arrives

VOUCH Theorem 5 bounds the certification time only through refutation, which fires late. Here we bound it through wealth growth. The key observation is that under a uniform reveal order the betting edge of the without-replacement process is *deterministic*.

Lemma 11 (Deterministic edge)**.** *Fix the upper certificate arm with boundary $p_0 = \frac{1}{2} + \frac{\varepsilon}{2}$. Let the cohort carry j ones, $\hat{p} = j/m < p_0$, and $\delta = p_0 - \hat{p}$. Under a uniform reveal order, the recentred boundary $p_0(t)$ of VOUCH Theorem 2 and the conditional mean $r_t = \mathbb{E}[Z_{\pi(t)} \mid \mathcal{F}_{t-1}]$ satisfy, for every t and every history,*

$$g_t \;:=\; p_0(t) - r_t \;=\; \frac{m\delta}{m - t + 1}.$$

Proof. With S_{t-1} ones revealed, $p_0(t) = (mp_0 - S_{t-1})/(m - t + 1)$ and $r_t = (j - S_{t-1})/(m - t + 1)$; subtract. □

The edge grows as the cohort is exhausted, which is the finite-population advantage over an i.i.d. stream with the same mean, where the edge stays δ.

Theorem 12 (Certification time through wealth growth)**.** *Run the upper arm with a uniform mixture of K constant stakes $\Lambda = \{\lambda_1, \dots, \lambda_K\} \subset (0, \frac{1}{2}]$, refuting the null outright if $p_0(t) > 1$. Let $A_t = \sum_{u \le t} g_u = m\delta\,(H_m - H_{m-t})$ and $G_t = \sum_{u \le t} g_u^2 = m^2\delta^2 \sum_{k=m-t+1}^{m} k^{-2}$, with H_n the harmonic numbers, and for $\lambda \in \Lambda$*

$$\Psi_t(\lambda) = \lambda A_t - \frac{\lambda^2(t/4 + G_t)}{2(1-\lambda)^2}, \quad V_t(\lambda) = \frac{\lambda^2(t/4 + G_t)}{(1-\lambda)^2}, \quad R_\lambda = \log\frac{1+\lambda}{1-\lambda}.$$

Then the stopping time τ of the arm (reaching $1/\alpha$ or refuting) satisfies, for every $t < m$,

$$\Pr[\tau > t] \;\le\; \min_{\lambda \in \Lambda} \exp\Big(-\frac{x_t(\lambda)^2}{2\big(V_t(\lambda) + R_\lambda x_t(\lambda)/3\big)}\Big), \qquad x_t(\lambda) = \big(\Psi_t(\lambda) - \log(K/\alpha)\big)_+.$$

Proof. Fix $\lambda \in \Lambda$ and write $X_u = \log\big(1 + \lambda(p_0(u) - Z_{\pi(u)})\big)$. Before refutation $p_0(u) \in [0,1]$, so $y_u = \lambda(p_0(u) - Z_{\pi(u)}) \in [-\lambda, \lambda]$. By Taylor's theorem with remainder, $\log(1+y) \ge y - y^2/(2(1-\lambda)^2)$ for $y \ge -\lambda$, hence $\mathbb{E}[X_u \mid \mathcal{F}_{u-1}] \ge \lambda g_u - \lambda^2 \mathbb{E}[(p_0(u) - Z)^2 \mid \mathcal{F}_{u-1}]/(2(1-\lambda)^2)$, and $\mathbb{E}[(p_0(u) - Z)^2 \mid \mathcal{F}_{u-1}] = r_u(1 - r_u) + g_u^2 \le \frac{1}{4} + g_u^2$. Summing, the compensator of $\log W_t^\lambda$ is at least $\Psi_t(\lambda)$, a deterministic quantity by Lemma 11. Also $|\log(1+y)| \le |y|/(1-\lambda)$, so the conditional variances sum to at most $V_t(\lambda)$, and the centred increments are bounded below by $-R_\lambda$. After refutation, replace X_u by its compensator lower bound so that the centred sequence stays a martingale difference sequence; this changes nothing on the event that refutation has not occurred by t. Freedman's inequality then gives $\Pr[\log W_t^\lambda < \Psi_t(\lambda) - x] \le \exp(-x^2/(2(V_t + R_\lambda x/3)))$. Finally, the mixture wealth is at least W_t^λ/K, so $\{\tau > t\} \subseteq \{\log W_t^\lambda < \log(K/\alpha)\}$ on the event of no refutation by t; take $x = x_t(\lambda)$ and minimise over Λ. □

Remark 13 (Reading the bound). Since $1/(m-u+1) \ge 1/m$, $A_t \ge \delta t$, with equality only for an infinite cohort; the factor $\rho_t = m(H_m - H_{m-t})/t \approx 1 + t/(2m)$ is the finite-population speed-up. Ignoring G_t and the factor $(1-\lambda)^{-2}$ and optimising λ, the drift is about $2\delta^2 t\rho_t^2$, so the time scale is $t^* \approx \log(K/\alpha)/(2\delta^2\rho_{t^*}^2)$, against $\log(1/\alpha)/\mathrm{KL} \approx \log(1/\alpha)/(2\delta^2)$ for an i.i.d. stream. This explains the VOUCH observation that the cohort column of its power table sits below the super-population information limit: the cohort answers an easier question, and the edge grows as the cohort runs out. The lower arm is the mirror statement with $\delta' = \hat{p} - (1 - p_0)$, and the two-sided certificate needs both arms (union bound).

Remark 14 (Numerical check). `extension/experiments/check_cohort_growth.py` compares the bound with the empirical tail of the same mixture bettor over 2,000 random orders for $(m, \varepsilon) \in \{(512, 0.2), (1024, 0.2), (1024, 0.1), (2048, 0.1)\}$ at $\hat{p} = \frac{1}{2}$. The bound is never violated at any of the 32 checked times per setting. It is conservative: the time at which it falls below one half is 2.2 to 3.8 times the empirical median. The same bettor against the fixed boundary on an i.i.d. stream with the same mean is slower, for example a median of 527 against 370 pairs at $m = 1024$, $\varepsilon = 0.1$.

6 Experiments

The experiments test two claims of this note on real language models: that Theorem 8 has teeth, in the sense that canary designs differ sharply in how detectable they are and the gap can be closed by construction, and that semantic twins with paraphrase scores (Proposition 2) see knowledge that a literal-form audit misses after partial unlearning. Every number below is produced by a script under `extension/experiments/` and stored under `extension/results/`; a test suite (`extension/tests/`, 11 tests) checks the generators and the exact null without model weights.

6.1 Setup

Data and models. The organic records are fictitious biographies drawn from six sentence templates (birthplace, study city, publication city, years before a profession, a guild prize, a workshop) over fictitious names, cities and professions, the keep/forget data of the VOUCH experiments. The main model is Pythia-410M; GPT-2 (124M) is used for the detectability features and for a pilot. A LoRA adapter (rank 16, all linear layers) is fine-tuned for 600 steps (batch 8, learning rate $2 \cdot 10^{-4}$, blocks of 96 tokens) on 2,000 keep records, 200 forget records and the in-twins of $m = 384$ semantic pairs, repeated $1, 2, 4$ or 8 times by stratum.

Semantic twins. Each pair names a fresh entity and one attribute (birthplace, profession, instrument, pet or colour) with two values drawn as an unordered pair and swapped by a fair coin, so the generator is label-symmetric. The in-twin is trained in its literal form ("X was born in Y.") and in one question-answer form. Three scores compare the two values under the same prompt: *literal* (mean token log-probability after the literal prefix), *belief* (the same, averaged over four paraphrased questions that never appear in training) and *belief_max*. The suffix *cal* subtracts the frozen base model's score difference, which removes its prior preference between the two values. Verdicts use the VOUCH engine at $\alpha = 0.05$ with a locally drawn reveal order; "literal" runs the engine on the literal score alone, "paraphrase" on belief_cal and belief_max_cal together. Recall that REVOKED rejects the exact null (some dependence on the coins) at any ε, while ISSUED certifies $|\Delta| < \varepsilon$.

Unlearning. Gradient ascent (GA) and NPO act on a copy of the fine-tuned adapter (batch 4, learning rate 10^{-4}; NPO with a retain term on the keep data and the fine-tuned adapter as reference). Retraining is a fresh adapter trained on the keep data only, the negative control; no unlearning is the positive control.

6.2 Canary detectability (Theorem 8)

Theorem 8 bounds the gap between canary and organic advantage by twice the distinguishing advantage γ, which data can only bound from below. We train two detectors to tell 400 canary in-twins from 400 organic records: a logistic regression on GPT-2 perplexity features (mean, maximum, median and spread of the token losses, and length) and a logistic regression on character 2–4-grams, both scored out of fold (5 folds). A detector's advantage $\max_\tau |\mathrm{TPR} - \mathrm{FPR}|$ is a lower bound on γ. For reference, a detector whose scores are independent of the label reaches 0.095 and 0.115 at the 95th and 99th percentiles at this sample size (4,000 simulations).

Four designs are compared. *vouch_qa* is VOUCH's random-string question-answer canary. *semantic_literal* and *semantic_qa* are the semantic twins above. *matched* draws each twin from the organic record generator itself (same templates and frequencies, plain names, same value draws) and varies a single slot, so that the in-twin is distributed as an organic record and $\gamma = 0$ at the record level by construction (up to names drawn without replacement).

Table 1 shows that every design built for VOUCH is detectable with advantage one: random strings, middle-initial names and attributes that never occur in organic data give canaries away to the simplest detector. For those designs Corollary 10 carries no information about organic records. The matched design brings both detectors to chance. Its perplexity advantage is below the 95th percentile of the null reference and its n-gram advantage lies between the 95th and 99th; the n-gram AUC is below one half, so the detector does not separate the two classes in the direction it was trained for and the advantage comes from the absolute value in its definition. These numbers are

Table 1: Detecting canaries among organic records ($n = 400 + 400$, out of fold). Advantage is a lower bound on γ.

	Perplexity features		Character n-grams		
Design	AUC	advantage	AUC	advantage	$\gamma \geq$
vouch_qa	0.998	0.973	1.000	1.000	1.000
semantic_literal	0.998	0.978	1.000	1.000	1.000
semantic_qa	0.892	0.605	1.000	1.000	1.000
matched	0.521	0.052	0.442	0.102	0.102

lower bounds on γ against two detector families; they do not show that no detector exists, which is the conditional nature of Theorem 8 noted in its remark.

6.3 Auditing with semantic twins

Table 2 reports the advantages and the two-sided verdict at $\varepsilon = 0.1$ over all scores, three seeds, when the forget set contains every form the in-twins were trained in.

Table 2: Semantic-twin audit on Pythia-410M ($m = 384$, three seeds). Advantages per seed; verdict over all scores at $\varepsilon = 0.1$.

Subject	literal	paraphrase (cal)	verdict
no unlearning	$+0.29,\ +0.32,\ +0.46$	$+0.20,\ +0.27,\ +0.26$	REVOKED $\times 3$
GA (100 steps)	$+0.03,\ +0.07,\ -0.05$	$+0.05,\ +0.05,\ -0.07$	ISSUED $\times 3$
NPO (250 steps)	$+0.03,\ -0.09,\ -0.04$	$+0.07,\ -0.07,\ -0.03$	ISSUED $\times 3$
retrain	$-0.03,\ -0.05,\ +0.03$	$+0.01,\ +0.12,\ +0.08$	ISSUED, UNDET., ISSUED

Paraphrased questions that were never trained carry about two thirds of the literal advantage, and the audit separates the positive from the negative control in every seed. A pilot with GPT-2 (128 pairs, 400 steps) did not: its positive control showed advantages of +0.25 (literal) and +0.17 (paraphrase) and received no revocation, so semantic twins need a model that generalises a stated fact to its paraphrases, which the smaller model barely does. With the full forget set, GA and NPO remove both the literal and the paraphrase advantage, and nothing distinguishes the two scores.

6.4 Partial unlearning and the squeezing effect

The case a literal audit cannot see is unlearning that removes the trained surface form but not the fact. To provoke it, the forget set is restricted to the literal statements (the question-answer form stays out of it) and the strength of unlearning is swept: GA for 5–25 steps, NPO for 10–75 steps, eight seeds each, and in three of the seeds also GA for 50 and 100 steps and NPO for 250. Define the *retention* of a score as its advantage after unlearning divided by the same seed's advantage before unlearning.

Two findings come out of Figure 1 and Table 3.

The literal form is unlearned faster than the fact. At every strength where both advantages are still clearly non-zero, paraphrase retention exceeds literal retention. Pooled over GA 10–20 and NPO 25–40 (40 seed-level cells), the mean difference is 0.21 ± 0.03, positive in 34 cells, Wilcoxon $p = 2 \cdot 10^{-7}$. Before unlearning the paraphrase advantage is about 60% of the literal one; after 15

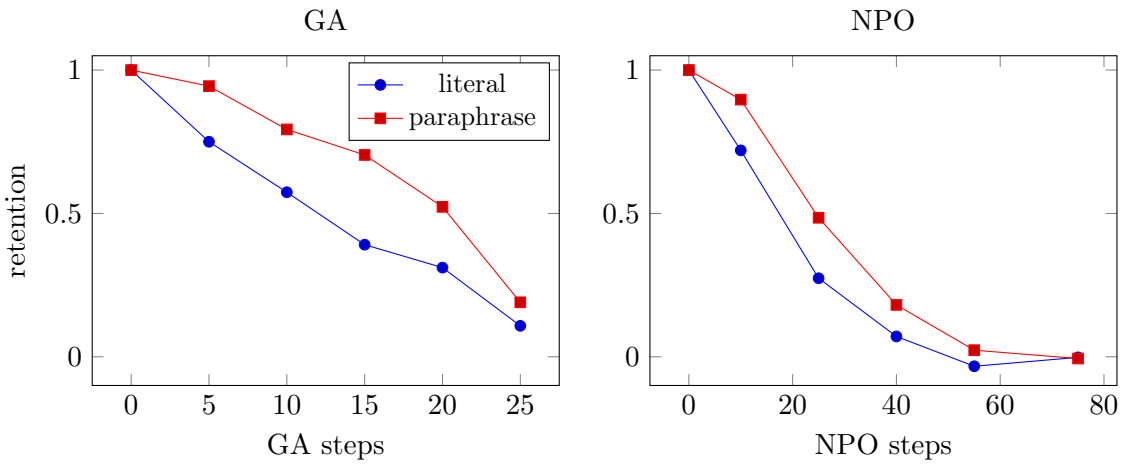


Figure 1: Retention of the literal and the calibrated paraphrase advantage under literal-only unlearning, mean over eight seeds (Pythia-410M, $m = 384$).

Table 3: Literal-only unlearning, eight seeds. Advantages are means ± standard errors; p is the Wilcoxon signed-rank test of paraphrase minus literal retention. The last two columns count seeds in which the two single-score audits disagree at $\varepsilon = 0.2$: A, literal ISSUED and paraphrase REVOKED; B, the reverse.

Subject	literal	paraphrase (cal)	retention lit. / para.	p	A	B
none	$+0.363 \pm 0.026$	$+0.219 \pm 0.016$	1.00 / 1.00	–	0	1
GA 5	$+0.275 \pm 0.028$	$+0.207 \pm 0.019$	0.75 / 0.94	0.039	0	1
GA 10	$+0.210 \pm 0.024$	$+0.175 \pm 0.020$	0.57 / 0.79	0.039	1	1
GA 15	$+0.143 \pm 0.016$	$+0.157 \pm 0.020$	0.39 / 0.70	0.008	1	1
GA 20	$+0.113 \pm 0.023$	$+0.120 \pm 0.019$	0.31 / 0.52	0.008	2	2
GA 25	$+0.036 \pm 0.015$	$+0.038 \pm 0.016$	0.11 / 0.19	0.250	0	1
NPO 10	$+0.259 \pm 0.019$	$+0.195 \pm 0.020$	0.72 / 0.90	0.039	0	0
NPO 25	$+0.097 \pm 0.015$	$+0.109 \pm 0.022$	0.27 / 0.49	0.055	3	1
NPO 40	$+0.020 \pm 0.027$	$+0.046 \pm 0.026$	0.07 / 0.18	0.109	2	0
NPO 55	-0.013 ± 0.014	$+0.007 \pm 0.028$	−0.03 / 0.02	0.742	1	0
NPO 75	-0.006 ± 0.017	$+0.005 \pm 0.021$	0.00 / −0.01	0.844	0	0

GA steps or 25 NPO steps the two are equal. A literal score therefore understates how much of the fact survives partial unlearning, by a factor close to two at NPO 25 (0.27 against 0.49).

At the level of verdicts the effect is small and partly noise. In the region GA 10–20 and NPO 25–55 (48 cells), the literal audit issues a certificate while the paraphrase audit revokes in 10 cells. The mirror image, a revocation by the literal audit and a certificate from the paraphrase audit, occurs in 5. Both kinds arise when two tests sit near their thresholds, so the 10 cells should not be read as ten misleading certificates; the excess of 5 over the mirror count is the part attributable to squeezing. The paraphrase audit is also the less powerful one: on the positive control it revokes in 7 of 8 seeds against 8 of 8 for the literal audit. Strong unlearning (GA ≥ 25, NPO ≥ 75) removes both advantages and the two audits agree.

The practical reading is that the paraphrase scores should be added to the declared class, not substituted for the literal score: the combined audit over all scores revokes whenever either score carries evidence, and its cost is a few extra forward passes per pair.

6.5 Certification time (Theorem 12)

The numerical check in the remark after Theorem 12 is the experiment for direction 3: 2,000 random orders in four settings, no violation of the bound at 128 checked times, the bound conservative by a factor of 2.2 to 3.8 at its median, and the without-replacement boundary faster than the fixed one in every setting.

7 What remains open

- Theorem 12 covers a constant-stake mixture, not the adaptive experts (ONS, KT) of the VOUCH engine. Including the constant-stake grid among the engine's experts inherits the bound at the cost of log of the expert count.
- Theorem 8 is conditional on γ, which data can only bound from below. Closing that gap needs either a computational assumption on the provider or a cryptographic attestation; neither is attempted here.
- The squeezing experiments use one model size and fictitious facts. Whether the retention gap widens with scale, and whether it persists for facts the model also knows from pre-training, is untested.
- The organic advantage Δ_ξ^o is a counterfactual quantity: organic records were not included by a coin. It is the advantage the pipeline *would* show on organic data under randomised inclusion, which is the standard definition of membership leakage, but not an observed rate.